

1.2 Binary Arithmetic

Binary Arithmetic operations

- Addition
- Subtraction
 - Direct method
 - 1's complement method
 - 2's complement method
- Multiplication
- Division

Binary Addition

Binary Addition

- When adding binary numbers we use binary logic
- Binary Addition Truth Table 1:

A	B	A + B	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Examples of binary addition

Example 1

Binary Addition - Examples

$$\begin{array}{r} \square \quad 101 \quad 5 \\ + 101 \quad + 5 \\ \hline 1010 \quad 10 \end{array}$$

$$\begin{array}{r} 00011010 + 26 \\ 00001100 \quad 12 \\ \hline 00100110 \quad 38 \end{array}$$

Binary addition

Example 2

		MSB			LSB	Decimal Equivalent
First Operand		1	1	1	1	15
Second Operand		1	0	1	0	10
Sum	1	1	0	0	1	25
Carry	1					

Binary addition

Example 3

$$\begin{array}{r} (85)_{10} \\ + (181)_{10} \\ \hline (266)_{10} \end{array} \quad + \quad \begin{array}{r} 11111 \\ 01010101 \\ 10110101 \\ \hline 100001010 \end{array}$$

Binary addition

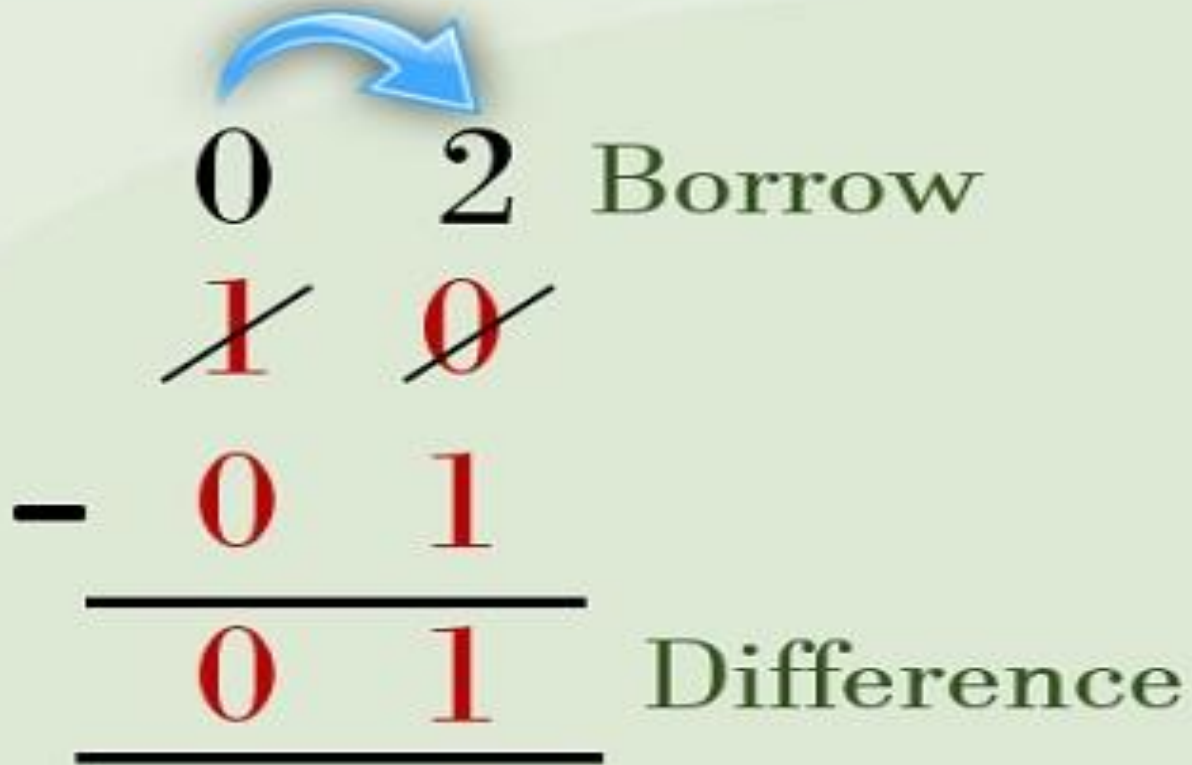
Example 4

$$\begin{array}{r} \overset{1}{1} \overset{1}{1} \overset{1}{1} \\ 1011.01 \\ + 11.011 \\ \hline 1110.101 \end{array}$$

Binary Subtraction- Rules

Inputs		Outputs	
Minuend <i>A</i>	Subtrahend <i>B</i>	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0
<i>A - B</i>		<i>Di</i>	<i>Bo</i>

Binary Subtraction-Examples



The diagram illustrates a binary subtraction step. At the top, a blue curved arrow points from the '2' to the '0' above it, indicating a borrow. Below this, the number '0 2' is shown, followed by the word 'Borrow'. Underneath, the digits '1' and '0' are crossed out with red diagonal lines. Below these, the digits '0' and '1' are shown, preceded by a minus sign. A horizontal line is drawn under these digits. Below the line, the digits '0' and '1' are shown, followed by the word 'Difference'. A second horizontal line is drawn under these digits.

$$\begin{array}{r} \text{0 2 Borrow} \\ \cancel{1} \cancel{0} \\ - \quad 0 \quad 1 \\ \hline 0 \quad 1 \text{ Difference} \\ \hline \end{array}$$

Binary subtraction –Example2

$$\begin{array}{r} (26)_{10} \\ -(12)_{10} \\ \hline (14)_{10} \end{array} \qquad \begin{array}{r} 11 \\ 00\cancel{1}\cancel{1}010 \\ - 0001100 \\ \hline 0001110 \end{array}$$

Binary subtraction –Example3

$$\begin{array}{r} (26)_{10} \\ -(12)_{10} \\ \hline (14)_{10} \end{array} \qquad \begin{array}{r} 11 \\ 00\cancel{1}\cancel{1}010 \\ - 0001100 \\ \hline 0001110 \end{array}$$

Binary subtraction –Example4

$$\begin{array}{r} \begin{array}{ccccccc} & 0 & & 0 & 1 & 1 & 0 & & 1 & 0 \\ & \swarrow & & \swarrow & \swarrow & \swarrow & & \swarrow & & \swarrow \\ \cancel{1} & 0 & \cancel{1} & 0 & \cancel{1} & 0 & 1 & . & \cancel{1} & 0 & 1 \end{array} \\ - \quad \quad 1011.11 \\ \hline 1001.111 \end{array}$$

Binary subtraction- 1's complement method

- $A - B$ where A is minuend and B is subtrahend
- This subtraction can be done as $A + (-B)$
- In 1's complement method subtrahend number is inverted and added to the minuend number.
- Bits 0's will be converted into bits 1's and bits 1's will be converted into bits 0's.

Examples of subtraction with 1's complement method

- 10110 – 00110
- Invert subtrahend number
- 00110 will become 11001 after inversion

$$\begin{array}{r} 10110 \\ + 11001 \\ \hline 101111 \end{array}$$

End around carry

Rules in 1's complement for End around carry bit

- Rule 1: If carry bit is generated in the MSB bits addition, it is end around carry. This indicates answer is positive and has to be added in LSB to get the correct answer.
- So, in previous example end around carry will be added in LSB as follows:

$$\begin{array}{r} 10111 \\ + \quad \quad \quad 1 \\ \hline 10000 \end{array}$$


Rules in 1's complement for End around carry bit contd...

- Rule 2: If end around carry bit is not generated end around carry, this indicates answer is negative. Answer is 1's complement and negative sign is put before the number.
- Example $10010 - 11010$

Take 1's complement of $11010 \rightarrow 00101$ and add in 10010

$$\begin{array}{r} 10010 \\ +00101 \\ \hline 1\ 01\ 1\ 1 \end{array}$$

- As there is no end carry around generated, take 1's complement of answer and put -ve sign

So for the above example answer is - 01000

Binary subtraction with 2's complement method

- 2's complement of binary number is obtained by adding 1 into its 1's complement number.

- Simple example:

$$7_{(10)} - 5_{(10)} = 2_{(10)}$$

- Binary subtraction with 2's complement:

$$0111 - 0101$$

- 1's complement of 0101 is 1010

- 2's complement will be $1010 + 0001 = 1011$

$$\begin{array}{r}
 0111 \\
 + 1011 \\
 \hline
 10010
 \end{array}$$

End around carry – Ignore it

So answer is $0010 = 2_{(10)}$

Example with –ve answer

- Now perform $5_{(10)} - 7_{(10)} = -2_{(10)}$
- $0101 - 0111$
- 1's complement of 0111 is 1000
- 2's complement of 0111 will be $1000 + 0001 = 1001$

$$\begin{array}{r} 0101 \\ + 1001 \\ \hline 1110 \end{array}$$

No end around carry is generated, so answer is negative and is in 2's complement form.

2's complement of $1110 = 0010$

So answer is $-2_{(10)}$

Subtraction with 2's complement method example 3

GROW

Example 1 =

$$(1001)_2 - (0101)_2$$

$$X = (1001)_2 \quad (9)_{10}$$

$$Y = (0101)_2 \quad (5)_{10}$$

Subtracted no. convert into 2's complement

$$\text{1's complement } Y = 1010$$

$$\text{2's complement } \begin{array}{r} +1 \\ \hline 1011 \end{array} \quad (-5)_{10}$$



Add X AND Y

$$\begin{array}{r} 1001 \quad (9)_{10} \\ + 1011 \quad (-5)_{10} \\ \hline 10100 \quad (20)_{10} \end{array}$$

If carry is generated result is +ve and true

If carry not produce result is -ve and 2's comp form

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Subtraction with 2's complement method

example 4

Decimal 2's Complement

17	0 0 0 1 0 0 0 1	→ Minuend
10 -	1 1 1 1 0 1 0 1	→ Subtrahend
	+ 1	→ Add 1
	(1) 1 1 1 0 0 0 1 0	Carry
7	0 0 0 0 0 1 1 1	→ Answer
	↓	
	Discarded	

Rules of end around carry in subtraction with 2's complement method

- If end around carry is generated it is ignored, answer is positive number
- If end around carry is not generated, answer is negative number represented in 2's complement form.

Two's Complement Overflow Rules

- The rules for detecting overflow in a two's complement sum are simple:
 1. If the sum of two positive numbers yields a negative result, the sum has overflowed.
 2. If the sum of two negative numbers yields a positive result, the sum has overflowed.
 3. Otherwise, the sum has not overflowed.

Comparison between 1's and 2's complements



1's Complement	2's Complement
It can be easily obtained using an Inverter	It has to be arrived at by first obtaining the 1's complement and then adding one (1) to it
It requires two operations	Only one arithmetic operation is required
It is often used in logical manipulations for inversion operation	It is used only for arithmetic applications

Binary Multiplication

0×0	0
0×1	0
1×0	0
1×1	1

Binary Multiplication Table

Binary multiplication example

$$\begin{array}{r} 1010 \\ \times 1001 \\ \hline 1010 \\ 0000 \\ 0000 \\ 1010 \\ \hline 1011010 \end{array}$$

Additive method of Binary Multiplication

- Most of the computers perform multiplication operation by the way of addition only. This can be easily seen by considering an example, say 5×8 . The result of this product can be calculated by evaluating $8 + 8 + 8 + 8 + 8$. The result is simply obtained by adding the digit 8 five times. Similarly computers perform all multiplication operation in binary using the additive approach.

Binary division

0 / 1	0
1 / 1	1

Binary Division Table

The rules for binary division :

- The division process is performed in a manner similar to decimal division.
- Start from the left of the dividend.
- Perform a series of subtraction in which the divisor is subtracted from the dividend.
- If subtraction is possible, put a 1 in the quotient and subtract the divisor from the corresponding digits of dividend.
- If subtraction is not possible (divisor is greater than remainder), record a 0 in the quotient.
- Bring down the next digit to add to the remainder digits. Proceed as before in a manner similar to long division.

Binary division example

$$\begin{array}{r} 0101 \\ 110 \overline{) 1000001} \\ \underline{110} \\ 1001 \\ \underline{110} \\ 11 \end{array}$$

- **Step 1:** Divisor greater than 100, so put 0 in quotient.
- **Step 2:** Add digit from dividend to group used above.
- **Step 3:** Subtraction possible so put 1 in the quotient.
- **Step 4:** Remainder from subtraction plus digit from dividend
- **Step 5:** Divisor greater, so put 0 in quotient.
- **Step 6:** Add digit from dividend to group.
- **Step 7:** Subtraction possible, so put 1 in quotient.

Octal Arithmetic

- Octal arithmetic operations are performed by converting octal numbers to binary numbers and then using rules of binary arithmetic
- In digital systems binary numbers required to be entered.
- It is highly inconvenient to handle long strings of binary numbers
- It may cause errors also. So octal numbers are used for entering the binary the binary data and displaying the information
- Therefore knowledge of octal numbers is important for efficient use of digital circuits

Example of octal addition

- $23_{(8)} + 67_{(8)}$

$$23_{(8)} = 010011$$

$$\underline{67_{(8)}} = \underline{1\ 10111}$$

$$112_{(8)} \quad 1001010$$

Method 2

Decimal Base-10	Binary Base-2	Octal Base-8	Hexa Decimal Base-16
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F
16	10000	20	10

Referring the table

- For $23_{(8)} + 67_{(8)}$
- $7+3 = 10$ in decimal. It's octal equivalent is 12
- $2+6 + \text{carry} = 9_{(10)}$. It's Octal equivalent is 11
- So answer is $112_{(8)}$

Octal subtraction

- $53_{(8)} = 00101011$

- $- 37_{(8)} = + 11100001$ 2's complement of $37_{(8)}$

$$\begin{array}{r} \hline (14)_8 \quad 100001100 \\ \hline \end{array}$$

Discard carry

- $(26)_8 = 00010110$

- $- (75)_8 = + 11000011$ 2's complement of $(75)_8$

$$\begin{array}{r} \hline -(47)_8 \quad 11011001 \quad \text{2's complement of result} \\ \hline \end{array}$$

Hexadecimal Arithmetic

- Hexadecimal number is converted into binary number and using binary arithmetic rules operations are performed
- Hexadecimal addition example

$$7F = 01111111$$

$$\begin{array}{r} +BA = 10111010 \\ \hline \end{array}$$

$$139_{(16)} \quad 100111001$$

Hexadecimal subtraction

$$1) \text{ CO} = 110000000$$

$$\underline{-7A} = + \underline{10000110} \text{ 2's complement of } 7A$$

$$46_{(16)} \quad 101000110$$

Discard carry

$$2) 3F = 00111111$$

$$\underline{-5C} = + \underline{10100100} \text{ 2's complement of } 5C$$

$$\underline{-1D} \quad 11100011 \text{ 2's complement of result}$$

Thank You !!